ENGINEERING RESEARCH DISSERTATION

PASSBRIDGE

A Finite-Conductivity Passive Microwave Reflector
for a Conditional 5.5 GHz Non-Line-of-Sight Link

A Coupled Electromagnetic–Structural–Communication Numerical Feasibility Study

Prepared by

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Research domains: Electromagnetics, microwave propagation, bistatic radar cross section, computational structural mechanics, digital communications, and systems engineering.

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Abstract

This dissertation develops and analyzes PASSBRIDGE, a stationary, energy-independent, galvanized-steel microwave reflector intended to redirect a radio path around an assumed obstruction. The reference configuration employs a 5.5 GHz carrier, two 500 m propagation segments, a nominal 90° change of direction, and a nearly 2 m × 2 m flat reflector assembled from two 999 mm × 2000 mm sheets separated by a 2 mm seam. Its electromagnetic model retains finite electrical conductivity in the steel substrate and zinc coating rather than adopting a perfect-electric-conductor surface. Its structural representation includes a perimeter rectangular-hollow-section frame, six internal stiffeners and four idealized support pads. A simplified uniform wind-pressure model is examined at 15 , 25 and 35 m/s using three-dimensional finite-element analysis (FEA).

The study couples a COMSOL Multiphysics structural displacement field to finite-thickness scattering geometry and evaluates the resulting bistatic radar cross section (RCS) using CST Studio Suite's high-frequency shooting-and-bouncing-rays (SBR) framework. A 5×5 angular perturbation grid spanning $\pm 0.2°$ on each axis is evaluated for each wind condition. The submitted dataset contains 75 completed geometric/tilt cases and 150 native TE/TM complex-field records. In the documented conditional link budget, with 29 dBm equivalent isotropically radiated power (EIRP), 23 dBi receiving gain and 6 dB additional loss, the smallest calculated received power is −75.9011 dBm. The −77 dBm project screening level is met by 1.099 dB in this worst simulated case. For 20 MHz receiver bandwidth and 7 dB noise figure, the corresponding calculated SNR is approximately 18.09 dB.

A separate MATLAB baseband experiment employs 64-point, uncoded QPSK-OFDM with 52 occupied carriers and a 16-sample cyclic prefix; its nominal payload-symbol rate is 26 Mbit/s. The 12 reflector-present communication conditions contain 200 packet trials each and show zero observed packet errors under a frequency-flat additive-white-Gaussian-noise (AWGN) channel assumption. This is not a prediction of Wi-Fi medium-access-control throughput or terrestrial link availability. The structural data present a material limitation: the 35 m/s peak displacement changes from 0.293 61 mm on one mesh to 0.833 23 mm on a finer mesh, and several recorded minimum element qualities are low. Hence structural deformation values are mesh-dependent and the exact RF consequences of a converged physical deformation are not yet established. The final evidentiary conclusion is numerical feasibility under a declared, simplified propagation model, not experimental proof or construction certification.

Index Terms— Passive microwave reflector, 5.5 GHz, non-line-of-sight (NLOS) link, galvanized steel, bistatic radar cross section, shooting and bouncing rays, FEM, wind loading, QPSK-OFDM, numerical feasibility.

Research Integrity and Scope

All numerical results presented as *computed* in this dissertation are traceable to the PASSBRIDGE project archive delivered on 27 October 2026, including native complex CST angular-field files, COMSOL displacement exports, project scripts and MATLAB result tables. Derived quantities, reformatted charts and analytical cross-checks are explicitly identified. Project software files document an automated solver workflow; the report presents the corresponding engineering modeling decisions without misrepresenting automated steps as exclusively manual GUI operations. Source-generated CAD schematics are labeled as such, and graphs derived from solver exports are not represented as photographic evidence. The reported simulation was not accompanied by construction, material coupon characterization, field-radio measurements, independent full-wave method-of-moments verification or certified structural assessment. This separation between methods, results, assumptions and inference follows established principles of reproducible numerical research [18, 19].

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Nomenclature

Symbol / Term	Definition
A	Geometric reflector area, m^2.
A_{proj}	Projected illuminated area, m^2.
B	Receiver noise-equivalent bandwidth, Hz.
C_D, C_f	Effective wind pressure/force coefficient (scenario value 1.8).
c	Speed of light in vacuum, m/s.
E, ν	Young's modulus and Poisson ratio in the mechanical model.
$\mathbf{E}, \mathbf{H}$	Electric and magnetic field phasors.
EIRP	Equivalent isotropically radiated power.
FEA/FEM	Finite-element analysis/method.
G_r	Receiver antenna gain, dimensionless or dBi.
$k_0 = 2\pi/\lambda$	Free-space wavenumber, rad/m.
L_{ext}	Additional, collectively budgeted RF loss, dB.
N_{FFT}, N_{CP}	OFDM transform size and cyclic-prefix length.
N_F	Receiver noise figure, dB.
P_r	Received power at the receiver reference point, dBm.
P_{ER}	Transmit EIRP, dBm.
R_1, R_2	Transmitter-to-reflector and reflector-to-receiver ranges, m.
RCS, σ_b	Bistatic radar cross section, m^2; $\sigma_{\text{dBsm}} = 10\log_{10}(\sigma_b/m^2)$.
RHS	Rectangular hollow section, used for steel framing.
SBR	Shooting-and-bouncing-rays high-frequency solver approximation.
SNR	Signal-to-noise ratio.
$\mathbf{u} = (u, v, w)$	Three-dimensional structural displacement vector, m.
V	Reference wind speed, m/s.
ϵ, μ, σ_e	Electric permittivity, magnetic permeability and electrical conductivity.
λ	Free-space wavelength at the carrier, m.
ρ_{air}	Air density, kg/m^3.
σ_{vm}	von Mises equivalent stress, Pa.
TE/TM	Transverse-electric / transverse-magnetic polarization conventions of the project export.

CHAPTER 1

Introduction and Research Definition

1.1 Engineering Motivation

Conventional terrestrial links depend on a sufficiently unobstructed direct propagation path or on engineered means of circumventing obstructions, including active repeaters, additional relay stations, fiber backhaul or alternate frequencies. In rural corridors, valleys, industrial campuses and installations with limited energy supply, the acquisition and operation of an active relay can become disproportionate to the desired service footprint. A passive reflector provides a different mechanism: an appropriately oriented electrically conducting aperture redirects a fraction of an illuminating field toward a second terminal. No power amplifier, local transceiver, frequency conversion or supply is incorporated in the reflective panel itself. The transmitter and receiver remain active devices. This distinction is central to the energy argument and the electromagnetic formulation [3, 2, 7].

For a practical candidate, a simple optical mirror analogy is inadequate. Microwave operation entails finite-aperture diffraction, phase-sensitive coherent scattering, surface resistance, polarization, imperfect alignment, seam discontinuities, structural deformation, and atmospheric and antenna-specific link losses. A nominal metallic plate may meet an analytical RCS estimate while failing as a durable outdoor construction component. Conversely, a stable steel structure can be electrically ineffective when the incidence or observation geometry is wrong. PASSBRIDGE therefore treats electromagnetics, structural mechanics, geometry transfer and communication-system performance as a single, explicitly staged design problem.

1.2 Problem Statement

The investigated problem is whether a nominal 2 m class galvanized-steel passive aperture can support a conditional two-hop geometrical ray path at 5.5 GHz, given an assumed obstruction of the direct transmitter–receiver path. The two segments each have length 500 m; the incoming and outgoing propagation directions differ by 90°. The target angular attitude is the specular condition for a flat panel. The reference screening rule requires the calculated receiver input power to exceed −77 dBm under the specified equipment assumptions and the tested wind/tilt combinations.

This problem is intentionally narrower than a universal outdoor service claim. Terrain profiles, vegetation, diffraction around alternate objects, multipath, antenna radiation-pattern measurements, receiver implementation loss, hardware dynamic range, interference, link availability and legal installation conditions are not inputs to the submitted solver results. The distinction between a model condition and a deployment guarantee is respected throughout [8, 26].

1.3 Objectives and Research Questions

The work addresses five connected questions. First, what aperture area and metal surface are required to obtain appreciable bistatic RCS in the desired direction? Second, what commercially realistic metallic surface can provide electrical conductivity while mitigating ordinary atmospheric corrosion? Third, what static deflections occur under the selected uniform wind pressures, and how reliably do the finite-element outputs converge? Fourth, how does the exported deformed geometry alter the TE/TM bistatic scattering when both mounting-axis angles vary within ±0.2°? Fifth, what receiver power, SNR and simplified uncoded QPSK-OFDM performance follow from the scattered-field records?

1.4 Principal Contributions

The dissertation brings together (i) a dimensioned two-sheet, two-layer-conductor panel and framed mounting concept; (ii) an explicit finite-conductivity electromagnetic treatment at the surface, rather than an ideal PEC shortcut; (iii) full-size static structural cases at three loads and four stored mesh levels; (iv) 3D displacement-to-surface transfer into finite-thickness RF geometry; (v) 150 retained native complex field cuts underlying an auditable 75-case angle–wind design; and (vi) a parametric power/SNR-to-OFDM assessment. The contribution is the engineering traceability and cross-disciplinary coupling of these models, rather than a claim of discovering a new scattering law or a new communication waveform.

1.5 Evidence Taxonomy

Four kinds of statements are kept distinct. *Model inputs* are explicitly selected dimensions, material constants, boundaries and receiver assumptions. *Solver outputs* are values stored by the COMSOL/CST/MATLAB campaign. *Analytical checks* are independently recomputed from governing equations and native output tables. *Engineering inferences* evaluate whether the combination plausibly meets the declared scenario. This classification prevents a case-dependent RCS or finite-packet observation from being overstated as a universal measured performance characteristic.

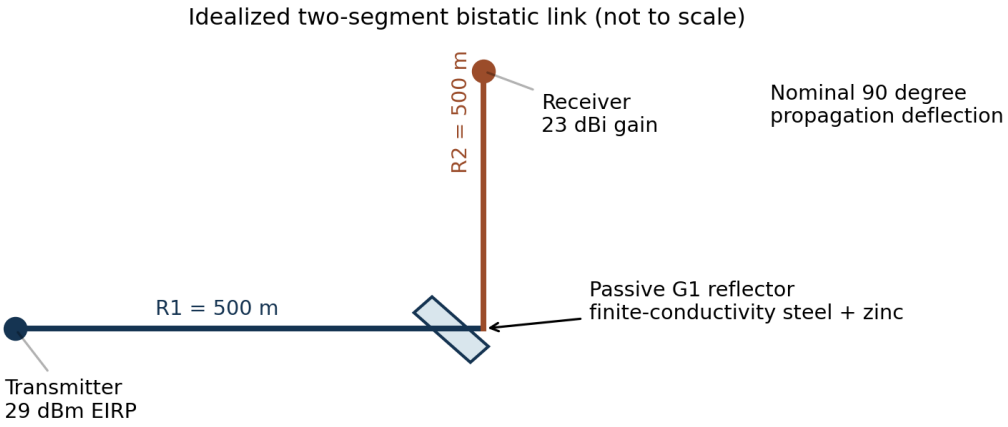


Figure 1.1: Idealized PASSBRIDGE two-segment propagation topology. Dimensions and terminal gains reflect declared model inputs. This is a schematic, not a surveyed path or site photograph.

1.6 Document Organization

Chapter 2 presents the scattering and link-budget theory. Chapter 3 establishes the geometry, material designation and conductor assumptions. Chapter 4 describes the structural FEM with mesh-dependent outputs. Chapter 5 specifies the transfer of genuine FEM fields into the RF shape. Chapter 6 documents SBR extraction and alignment experiments. Chapter 7 develops the link budget and sensitivity measures. Chapter 8 presents the OFDM signal model and statistics. Chapters 9–11 synthesize the evidence, limitations and engineering implications. Appendices provide parameter ledgers, complete summary results, reproducibility details and figure provenance.

CHAPTER 2

Electromagnetic and Propagation Theory

2.1 Maxwell Equations and the Conducting Interface

In an assumed linear isotropic material, the frequency-domain Maxwell curl relations are

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}, \tag{2.1}$$

$$\nabla \times \mathbf{H} = (\sigma_e + j\omega\epsilon)\mathbf{E}, \tag{2.2}$$

where the $e^{j\omega t}$ time convention is used. Tangential $\mathbf{E}$ and $\mathbf{H}$ satisfy the appropriate material boundary conditions, with conduction producing ohmic dissipation. A perfect electric conductor would enforce $\mathbf{E}_t = \mathbf{0}$ at the surface and imply vanishing resistive loss; PASSBRIDGE does not use this idealization for its modeled metal [1, 2].

For a good nonmagnetic conductor, the leading-order skin depth is

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma_e}} = \frac{1}{\sqrt{\pi f \mu \sigma_e}}. \tag{2.3}$$

The surface-impedance approximation is conventionally written

$$Z_s \simeq (1 + j)\sqrt{\frac{\omega\mu}{2\sigma_e}}, \qquad R_s = \mathrm{Re}\{Z_s\}, \tag{2.4}$$

subject to layer thickness and substrate effects. It is useful for explaining why a zinc-coated conductive panel can reflect strongly even when actual copper-like conductivity is not available. A multilayer finite-thickness model is necessary to capture the distinction between the coating and its substrate when conductivity or layer continuity differs materially [2, 1].

2.2 Reflection Geometry

Let $\hat{\mathbf{k}}_i$ be the incident propagation unit vector and $\hat{\mathbf{n}}$ the local outward surface normal. In the geometrical-optics limit the reflected direction is

$$\hat{\mathbf{k}}_r = \hat{\mathbf{k}}_i - 2(\hat{\mathbf{k}}_i \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}. \tag{2.5}$$

The right-hand side preserves the incident tangential wave-vector component and reverses its normal component. For a 90° deflection between incident and reflected propagation directions, the normal lies along the appropriate angular bisector and the nominal angle from the normal is 45°. A small change in mounting attitude changes the phase distribution across the entire aperture; because the panel is tens of wavelengths wide, a few tenths of a degree can have a meaningful effect on received scattering even if the mechanical change appears small.

2.3 Wavelength, Electrical Size and Fresnel Considerations

At the center frequency,

$$\lambda = \frac{c}{f} = \frac{299\,792\,458}{5.5 \times 10^9} = 0.054\,507\,7\ \mathrm{m}. \tag{2.6}$$

The 2 m side is approximately 36.69λ, so the panel is electrically large in both principal dimensions. The nominal distance that separates simple Fraunhofer far-field behavior from aperture near-field behavior is

$$R_{\mathrm{FF}} \approx \frac{2D^2}{\lambda}, \tag{2.7}$$

with D the largest aperture dimension. Taking $D = 2\,\mathrm{m}$ gives $R_{\mathrm{FF}} \approx 146.8\,\mathrm{m}$. Each modeled 500 m path lies beyond this approximate criterion for the panel alone, though a complete antenna-to-panel system must still consider actual feed aperture, taper and beam footprint [3, 2]. The approximation does not prove that arbitrary high-gain endpoint antennas fully illuminate or collect the required footprint.

2.4 Bistatic Radar Cross Section

The bistatic radar cross section compares scattered power-density amplitude at observation distance R with incident power density at the object:

$$\sigma_b(\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_s) = \lim_{R\to\infty} 4\pi R^2 \frac{|\mathbf{E}_s(R, \hat{\mathbf{k}}_s)|^2}{|\mathbf{E}_i|^2}. \tag{2.8}$$

It is an angular, frequency- and polarization-dependent quantity, not a dimensionless reflectivity or an antenna gain. σ_b has units of area, and its logarithmic representation is $\sigma_{\mathrm{dBsm}} = 10\log_{10}(\sigma_b/1\,\mathrm{m}^2)$. IEEE Std 145-2025 supplies a contemporary terminology framework for antenna-related quantities [6]; conventional scattering treatments are given in [3, 4, 5].

For an electrically large, perfectly conducting, planar rectangle near the specular direction, a useful physical-optics consistency expression is

$$\sigma_{\mathrm{PO,spec}} \approx \frac{4\pi A^2}{\lambda^2}\cos^2\theta_i, \tag{2.9}$$

where A is geometric panel area and θ_i denotes incidence with respect to the normal. This expression is a scaling check, not a substitute for the finite-conductivity, finite-seam numerical model; edge diffraction, finite illumination, polarization, finite resistivity and element geometry can move the response. Inserting $A \simeq 3.996\,\mathrm{m}^2$ and $\theta_i = 45^\circ$ yields approximately 45.3 dBsm, which is close to the documented nominal solver level. This agreement checks dimensional plausibility; it does not independently validate the full field solution.

2.5 Coherent Aperture Scattering and Phase Sensitivity

A physical-optics-type far-field integral schematically takes the form

$$\mathbf{E}_s(\hat{\mathbf{k}}_s) \propto \int_A \mathbf{J}_s(\mathbf{r})\, e^{jk_0(\hat{\mathbf{k}}_i - \hat{\mathbf{k}}_s)\cdot\mathbf{r}}\,\mathrm{d}A, \tag{2.10}$$

where the induced equivalent current $\mathbf{J}_s$ depends on the surface impedance, local illumination and polarization. Structural deformation changes $\mathbf{r}$ and the local normal. The resulting phase perturbation is, to first order,

$$\Delta\phi(\mathbf{r}) = k_0(\hat{\mathbf{k}}_i - \hat{\mathbf{k}}_s)\cdot\mathbf{u}(\mathbf{r}). \tag{2.11}$$

Equation (2.11) explains why a submillimeter deflection can matter at centimeter wavelengths and also why the spatial distribution of displacement matters more than a single maximum displacement statistic. A nearly rigid translation is physically different from a curved displacement field of the same peak amplitude.

2.6 Receiver Power for a Two-Segment Scattered Path

For an equivalent transmitter EIRP, a bistatic scatterer with cross section σ_b, and receiver gain G_r, the free-space power balance adopted in this study is

$$P_r = \mathrm{EIRP}\, G_r \frac{\lambda^2 \sigma_b}{(4\pi)^3 R_1^2 R_2^2} \frac{1}{L_{\mathrm{ext}}}, \tag{2.12}$$

with every power and loss factor expressed linearly. Expressed in dB units,

$$\begin{aligned} P_r\,[\mathrm{dBm}] = {} & P_{\mathrm{EIRP}}[\mathrm{dBm}] + G_r\,[\mathrm{dBi}] + \sigma_b\,[\mathrm{dBsm}] \\ & + 20\log_{10}\lambda - 30\log_{10}(4\pi) \\ & - 20\log_{10} R_1 - 20\log_{10} R_2 - L_{\mathrm{ext}}[\mathrm{dB}], \end{aligned} \tag{2.13}$$

where λ and both ranges are evaluated numerically in SI units before logarithms. This is a bistatic radar-equation-style link budget for a passive scatterer and must not be replaced by a 1000 m one-segment Friis formula [4, 7, 2].

2.7 Noise and Link Margin

The thermal-noise floor for a matched receiver model is

$$N[\mathrm{dBm}] = -174 + 10\log_{10}\left(\frac{B}{1\,\mathrm{Hz}}\right) + N_F\,[\mathrm{dB}], \tag{2.14}$$

using 290 K reference temperature. For $B = 20 \times 10^6$ Hz and $N_F = 7$ dB, this gives approximately −93.99 dBm. The link model then sets

$$\mathrm{SNR}_{\mathrm{dB}} = P_r\,[\mathrm{dBm}] - N[\mathrm{dBm}], \tag{2.15}$$

and defines the screening margin relative to an explicitly selected threshold P_{scr} as

$$M_{\mathrm{scr}} = P_r - P_{\mathrm{scr}}. \tag{2.16}$$

The screening margin is not automatically a fade margin, regulatory margin or full receiver-sensitivity reserve.

CHAPTER 3

Reflector Architecture and Material Engineering

3.1 Choice of Flat G1 Configuration

The numerical campaign converged on a flat two-sheet G1 configuration as the project reference. Previously considered faceted and curved alternatives were not carried into the archived 75-case coupled campaign, and no comparative G2/G3 native field datasets are included in the published 2026 GitHub archive. The design rationale is therefore articulated as a manageable aperture, comparatively simple fabrication and a direct relation between specular geometry and alignment. A universal ranking of alternative shapes is not claimed without equivalent published case data.

3.2 Dimensioned Geometric Definition

The front reflecting surface consists of two 999 mm × 2000 mm plates with a nominal 2 mm center seam. They span 2000 mm in width when placed with a common vertical extent. Their combined metal area is

$$A = 2(0.999 \times 2.000) = 3.996\,\mathrm{m}^2. \tag{3.1}$$

The model specifies a nominal 1.2 mm steel substrate thickness. The gap is a physical discontinuity, not a 2 mm thick additional metal strip. It can change coherent scattering locally and should not be deleted from a validation model without an explicit sensitivity study.

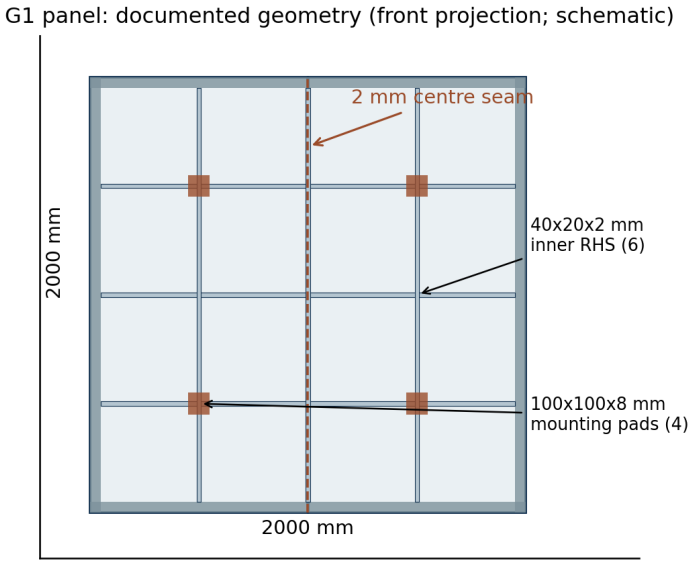


Figure 3.1: Front projection reconstructed from the recorded COMSOL geometry parameters: two reflecting sheets; perimeter frame; three horizontal plus three vertical internal rectangular-hollow sections; and four mounting pads. The drawing is a dimension-driven schematic, not a CST or COMSOL screenshot.

3.3 Why Galvanized Low-Carbon Steel?

A passive outdoor reflector requires a large continuous conductive surface but does not intrinsically need copper's maximal bulk conductivity. Ordinary galvanized low-carbon sheet is a viable engineering

Table 3.1: Documented G1 Geometry and Mechanical Support Parameters

Feature	Definition	Quantity
Conducting front sheets	$999 \times 2000 \times 1.2$ mm steel substrate	2
Central separation	Unfilled center seam, 2 mm	1
Perimeter RHS	$50 \times 50 \times 3$ mm, nominal 2 m members	4
Internal RHS	$40 \times 20 \times 2$ mm, nominal 1.9 m members	6
Mounting plates	$100 \times 100 \times 8$ mm idealized pads	4
Zn surface treatment	Z275 designation; nominal $\approx 19.26\,\mu$m/face	2 faces/sheet
Panel RF area	Two-sheet geometric area	$3.996\,\mathrm{m}^2$

candidate because it is comparatively accessible, mechanically formable, electrically conducting and supplied with sacrificial zinc corrosion protection. DX51D identifies low-carbon flat steel intended for cold forming, while Z275 denotes a nominal 275 g/m^2 total coating mass distributed over both sheet faces under the applicable coated-sheet specification. The coating designation does not, by itself, establish a guaranteed structural yield strength [9, 10, 11]. The project does not contain a purchase certificate, coating-thickness inspection or verified alloy lot data; the designation and numerical conductivities are therefore design assumptions rather than traceable material measurements.

The nominal per-face thickness of a uniform symmetric zinc coating can be estimated from the total areal mass:

$$t_{\mathrm{Zn}} = \frac{m_A}{2\rho_{\mathrm{Zn}}} = \frac{0.275\,\mathrm{kg\,m^{-2}}}{2(7140\,\mathrm{kg\,m^{-3}})} \approx 19.26\,\mu\mathrm{m}. \tag{3.2}$$

This value is consistent with the approximately 20 µm typical per-face coating reference in manufacturer information [10, 11]. It is not a guarantee that every location on either face has exactly this thickness.

3.4 Conductivity and Penetration Depth

The model uses provisional electrical conductivities of $\sigma_{e,\mathrm{steel}} = 6.0 \times 10^6\,\mathrm{S/m}$ and $\sigma_{e,\mathrm{Zn}} = 1.69 \times 10^7\,\mathrm{S/m}$, with $\mu_r = 1$ assumed. Inserting the zinc value in (2.3) gives a microwave skin depth of the order of 1.65 µm at 5.5 GHz. The nominal zinc coating therefore spans about 12 skin depths, making the outer layer electromagnetically consequential under a continuous-coating hypothesis. The finite-thickness layered representation nevertheless remains preferable because scratches, seam interfaces, edge contacts, conductivity dispersion and coating nonuniformity cannot be dismissed on the basis of an ideal homogeneous skin-depth calculation.

Table 3.2: Electrical and Structural Constitutive Inputs

Quantity	Model significance	Assigned value
Steel σ_e	Finite RF substrate conduction	$6.0 \times 10^6\,\mathrm{S/m}$
Zinc σ_e	Finite RF coating conduction	$1.69 \times 10^7\,\mathrm{S/m}$
μ_r	Simplified RF permeability	1
Young's modulus E	Linear elastic isotropic FEM input	210 GPa
Poisson's ratio ν	Linear elastic FEM input	0.30
Steel density ρ_s	Nominal FEM mass density	$7850\,\mathrm{kg/m^3}$
Zn density ρ_{Zn}	Coating-thickness estimate	$7140\,\mathrm{kg/m^3}$

3.5 Mass and Practical Construction Implications

The mass of the steel reflecting skins alone is approximately

$$m_{\text{steel}} = \rho_s A t_s = 7850 \times 3.996 \times 0.0012 \approx 37.64\,\text{kg}. \tag{3.3}$$

The ideal total zinc mass is $m_{\text{Zn}} = A(0.275) \approx 1.10\,\text{kg}$, giving a rough sheet-and-coating mass of 38.74 kg before mounting frame, stiffeners, welds, fasteners and mast attachments. These values are analytical estimates from nominal dimensions, not weighed prototype data. In particular, the support structure adds significant mass and wind load, so total assembly weight should not be reported as the sheet-only figure.

3.6 Corrosion, Fabrication and Alternatives

Zinc provides sacrificial corrosion protection and may simplify maintenance compared with uncoated carbon steel, although lifecycle behavior depends on atmospheric corrosivity, condensation, dissimilar-metal contact, scratches and edge finishing [12, 10]. Aluminium would lower density but entails its own alloy, stiffness, joint and surface-treatment tradeoffs; stainless steel can improve corrosion resistance at different material cost and conductivity. A copper sheet would increase bulk conductivity but impose material cost, oxidation and structural choices which are not justified solely by the near-specular scattering objective. PASSBRIDGE does not include procurement quotations or lifecycle-cost measurements, so its affordability is a rationale, not a measured economic comparison.

An assembly design would also have to state coating treatment at weld locations, galvanic compatibility of bolts, drainage and ventilation, tolerances on panel planarity, accessible adjustment points, and a method for measuring the true orientation of the assembled aperture. None of these fabrication and inspection records are present in the digital archive; they remain requirements for a later physical implementation.

CHAPTER 4

Structural Mechanics and Finite-Element Study

4.1 Structural Loading Concept

The reflector is a large, slender, plate-and-frame assembly. Structural response affects two performance criteria simultaneously: adequate support against imposed actions and retention of an electromagnetically useful surface figure. PASSBRIDGE represents wind as uniform static pressure on the nominal front metal surface. This is a parameterized numerical load case rather than a location-specific atmospheric-wind design performed according to building regulations. Real installations would require terrain-dependent exposure, turbulence, directional shape coefficients, transient gust analysis, fatigue, fastener checks and a site-specific governing design standard [13, 14].

A conventional reference dynamic pressure is

$$q = \tfrac{1}{2}\rho_{\text{air}} V^2. \tag{4.1}$$

The study multiplies by an assigned force coefficient $C_f = 1.8$:

$$p(V) = \tfrac{1}{2}\rho_{\text{air}} C_f V^2, \quad \rho_{\text{air}} = 1.225\,\text{kg/m}^3. \tag{4.2}$$

Thus $p = 1.1025V^2$ in pascals when V is in meters per second. The results are shown in Table 4.1. The idealized pressure-area product $F = pA$ indicates the total load transferred by the front panel, omitting shielding, frame wind exposure and nonuniform pressure fields. The quadratic dependence on wind speed provides a useful scaling check against the linear elastic simulations.

Table 4.1: Uniform Pressure and Nominal Net Sheet Force

Wind speed (m/s)	q (Pa)	$p = C_f q$ (Pa)	pA (kN)
15	137.813	248.063	0.991
25	382.813	689.063	2.754
35	750.313	1350.563	5.397

4.2 Continuum Formulation

For the small-displacement, small-strain, isotropic linear-elastic approximation, the symmetric strain tensor is

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \tfrac{1}{2}\left[\nabla\mathbf{u} + (\nabla\mathbf{u})^T\right]. \tag{4.3}$$

The Cauchy stress tensor is

$$\boldsymbol{\sigma} = \lambda_L \operatorname{tr}(\boldsymbol{\varepsilon})\mathbf{I} + 2G\boldsymbol{\varepsilon}, \tag{4.4}$$

where the Lamé coefficients are $G = E/[2(1+\nu)]$ and $\lambda_L = E\nu/[(1+\nu)(1-2\nu)]$. Static equilibrium, in the absence of the unmodeled body force terms, is

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0}. \tag{4.5}$$

On ideal supports the displacement is prescribed, whereas loaded boundaries satisfy $\boldsymbol{\sigma}\mathbf{n} = \mathbf{t}$, with $\mathbf{t}$ determined by the pressure convention. In weak form, the virtual-work statement is

$$\int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v})\, \mathrm{d}\Omega = \int_{\Omega} \mathbf{b} \cdot \mathbf{v}\, \mathrm{d}\Omega + \int_{\Gamma_t} \mathbf{t} \cdot \mathbf{v}\, \mathrm{d}\Gamma, \tag{4.6}$$

for admissible virtual displacement $\mathbf{v}$. Discretization by interpolation functions yields a linearized assembled system $\mathbf{Kd} = \mathbf{f}$. The reported maximum equivalent stress is the von Mises invariant

$$\sigma_{\mathrm{vm}} = \sqrt{\tfrac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}, \tag{4.7}$$

where σ_i are principal stresses [16, 17, 14].

4.3 Three-Dimensional Assembly and Support Representation

The source model defines two 1.2 mm thin steel plates; four $50 \times 50 \times 3$ mm perimeter RHS members; six $40 \times 20 \times 2$ mm intermediate RHS members; and four $100 \times 100 \times 8$ mm rear mounting plates. The RHS shapes are constructed from outer rectangular volumes with inner volumes removed, making the walls hollow rather than approximating each section as a solid steel bar. The joint idealization merges the modeled features through geometry union, corresponding to a rigidly bonded or welded conceptual connection in the continuum representation.

Four boxes identify the mounting-pad undersides, to which fixed conditions are applied. The front-face boundary selection establishes the pressure load. This is a deterministic reference support, not a calibrated mast, bolt preload, weld stiffness or anchor-to-ground measurement. A true structural qualification would require explicit checks for frame-joint stresses, buckling, connection fatigue, support movement, foundation reactions and wind-induced vibration.

4.4 Numerical Mesh Design and Output Extraction

The archived results include full-size solid-mechanics solutions at COMSOL automated mesh size levels 7, 6 and 5 and a later refined level 4. A lower automatic mesh-level number generally denotes a finer discretization in the software setup used here, but nominal level does not by itself quantify local resolution around each thin wall or sharply intersecting member. The model saved stress and displacement maxima for three wind pressures. Level 4 also exported genuine spatial nodal tables containing $x, y, z, u, v, w, \|\mathbf{u}\|$. The 35 m/s file identifies COMSOL 6.4.0.429, 50,807 exported nodes and four quantities; approximately 13,114 of those nodes lie on the nominal $z = 0$ front surface.

4.5 Computed Deformations and Stresses

The complete directly archived peak-value data are given in Table 4.2. In a linear elastic model with the same mesh and geometry, a constant change of load should scale both displacement and stress approximately with V^2. For mesh level 4, $0.83323/0.15304 \approx 5.44$, close to $(35/15)^2 \approx 5.44$. This load-scaling behavior supports internal linearity of the model; it does not establish correct boundary conditions or discretization independence.

4.6 Spatial Deformation Fields and Structural Interpretation

The detailed displacement files permit a stronger examination than only maximum values. Figure 4.2 displays the exported w component from the full-sized mesh-4 front face for the three wind cases. Its range becomes increasingly negative as pressure increases, while the pattern is constrained by the frame and rear mounting concept. Figure 4.3 magnifies the 35 m/s field without arbitrary displacement am-

Table 4.2: Archived Full-Panel FEA Responses at Four Mesh Levels

Wind (m/s)	Mesh level	Peak $\|\mathbf{u}\|$ (mm)	Peak σ_{vm} (MPa)	Min. quality
15	7	0.03803	5.2475	0.000011
15	6	0.05042	7.1203	0.003286
15	5	0.05393	6.9352	0.000414
15	4	0.15304	9.7202	0.004494
25	7	0.10565	14.5763	0.000011
25	6	0.14005	19.7785	0.003286
25	5	0.14980	19.2643	0.000414
25	4	0.42512	27.0007	0.004494
35	7	0.20707	28.5696	0.000011
35	6	0.27450	38.7659	0.003286
35	5	0.29361	37.7580	0.000414
35	4	0.83323	52.9213	0.004494

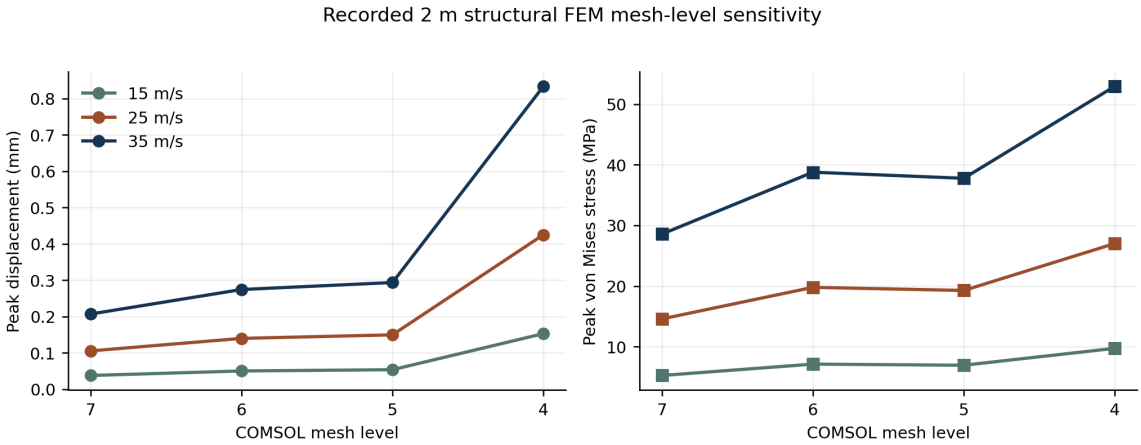


Figure 4.1: Displacement (left) and maximum von Mises stress (right) extracted from the archived full-size COMSOL cases. Lines connect recorded mesh-level results; they do not imply convergence.

plification. Both figures are constructed numerically from *actual archived COMSOL nodal coordinates and the exported displacement component*; they are not photographs or simulated renderings generated for presentation.

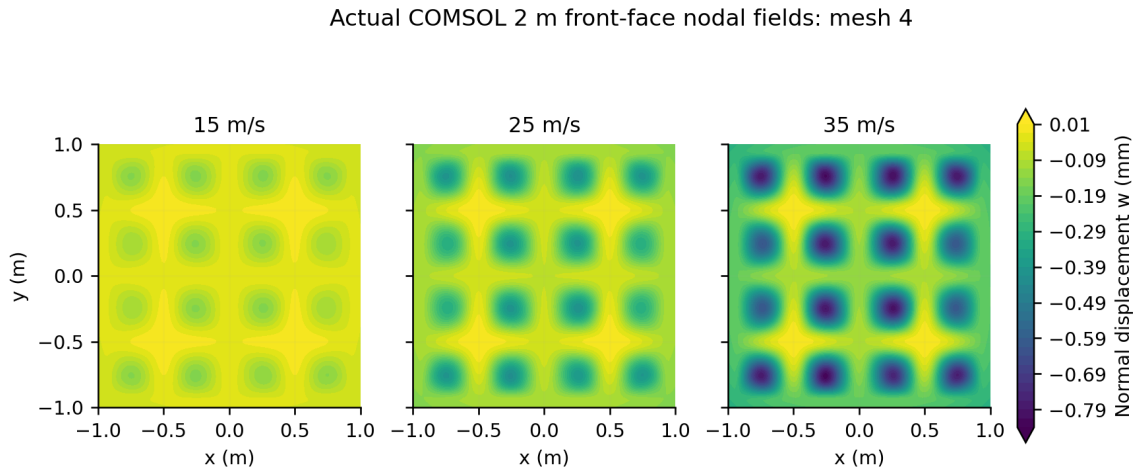


Figure 4.2: Genuine COMSOL mesh-4 front-surface displacement values for 15, 25 and 35 m/s; color encodes w in millimeters. Derived from the original three displacement export files.

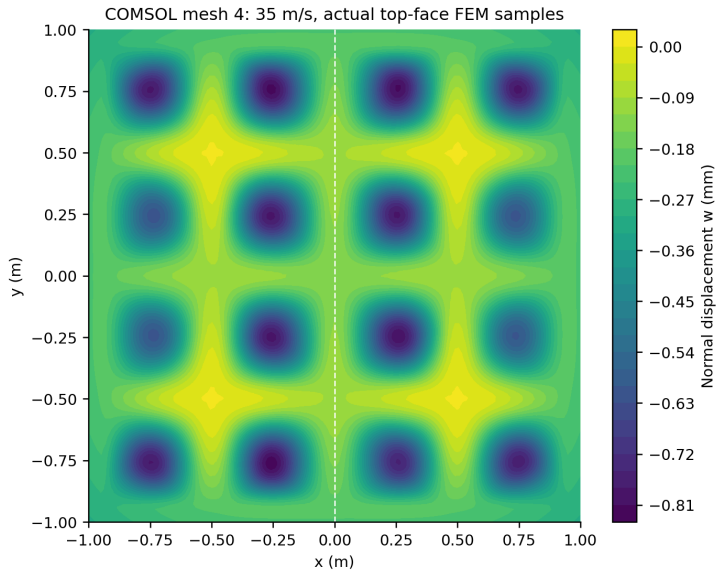


Figure 4.3: Detail of the actual 35 m/s front-face nodal field for the 2 m panel. The center seam is indicated by a dashed line; no mesh interpolation has been passed off as an instrument measurement.

4.7 Mesh-Quality and Convergence Assessment

Numerical convergence is a property to be demonstrated, not inferred from a solver's successful termination. Define a consecutive-level difference for an output Q by

$$\eta_Q = 100\% \times \frac{|Q_{h_2} - Q_{h_1}|}{|Q_{h_2}|}, \tag{4.8}$$

where h_2 denotes the more refined of the two compared discretizations. At 35 m/s, mesh 5 gives $\max|\mathbf{u}| = 0.293\,61$ mm, whereas mesh 4 gives 0.833 23 mm. The normalized difference is approximately 64.76% and the amplitude ratio is about 2.84. The peak stress changes from 37.758 MPa to 52.921 MPa, about 28.65% on the same denominator convention. Similar percentages are recorded for the other two winds because the constitutive model is linear.

The minimum reported element qualities across the four mesh levels include 1.077×10^{-5}, 4.14×10^{-4} and 4.494×10^{-3}. Low-quality small tetrahedra and extreme aspect ratios around thin plates and intersecting RHS members are credible explanations for non-robust stress and displacement extrema; geometry/contact support treatment is also a possible contributing factor. These are diagnostic hypotheses, not established root causes. A Richardson-style convergence or grid-convergence index cannot be stated credibly without consistent asymptotic behavior and representative refinement ratios [18, 20, 15].

The difference is not a trivial scientific footnote. The RF geometry in the coupling campaign is based on the mesh-4 spatial field, and its precise deformation magnitude is therefore not a mesh-independent physical prediction. The analysis can legitimately report that the simulated RF model functions under *the imported mesh-4 shape*; it cannot certify that a fabricated loaded panel would assume exactly that shape.

4.8 Structural Strength Versus Structural Verification

The largest numerical equivalent stress, 52.92 MPa, is below many common structural steel yield strengths. Nevertheless, DX51D's forming designation does not establish a certified minimum yield stress, and the source does not include a validated joint model or design-code resistance check. A factor of safety would formally require a known characteristic resistance, relevant partial factors, correct load combinations and certified supports:

$$\mathrm{FoS} = \frac{\text{verified resistance}}{\text{consistent governing demand}}. \tag{4.9}$$

Numerical stress divided into an assumed universal steel yield strength would create false precision; no verified construction safety factor is assigned here. The computation is a structural performance screen under explicit loads, not an installation approval.

CHAPTER 5

Transfer of Structural Deformation into the RF Model

5.1 Multiphysics Coupling Strategy

The sequential model treats deformed metal geometry as the bridge between mechanical and electromagnetic analyses. Its essential chain is: define loads and restraints; compute displacement; extract u, v, w at geometric points; interpolate the field onto a regularized electromagnetic surface; construct closed conductor volumes; preserve the metal-layer/material associations; and evaluate scattering for specified illumination and receiver directions. There is no asserted two-way electromagnetic-to-structural load coupling, thermal expansion or time-domain structural dynamics. The electromagnetic solver therefore operates on quasi-static shape snapshots.

5.2 Deformation Map and Interpolation

For each undeformed point $\mathbf{r} = (x, y, z)$, the displaced position is

$$\mathbf{r}' = \mathbf{r} + \mathbf{u}(\mathbf{r}). \tag{5.1}$$

The original COMSOL field is irregularly sampled in three-dimensional solid geometry. When constructing a structured surface for RF import, the transfer method employs nearest-neighbor-supported inverse-distance weighting (IDW). For a target $\mathbf{x}$, a representative expression is

$$\widehat{u}_\alpha(\mathbf{x}) = \frac{\sum_{j\in\mathcal{N}(\mathbf{x})} w_j(\mathbf{x}) u_{\alpha,j}}{\sum_{j\in\mathcal{N}(\mathbf{x})} w_j(\mathbf{x})}, \quad w_j(\mathbf{x}) = \frac{1}{(\|\mathbf{x}-\mathbf{x}_j\| + \varepsilon)^p}, \tag{5.2}$$

for $\alpha \in \{x, y, z\}$, interpolation power $p > 0$ and a small numerical regularizer ε. The exact sampling layout is documented in the supplied `FEMDEN_STL.py`; Equation (5.2) expresses the underlying method without asserting a separate interpolation-error calibration. Each new sample inherits the FEM displacement field rather than a manually imposed synthetic bowing function.

5.3 Closed Finite-Thickness Conductor Construction

The transfer creates six STL parts for each wind case: steel core/front and zinc-associated conductor faces grouped by the panel's two sides and material-layer naming scheme. These surface meshes describe finite-thickness metal solids, not a zero-thickness PEC sheet. The body coordinate representation includes both in-plane components and normal deformation; it does not simply shift the entire panel by its average w. Separate components support assignment of provisional zinc and steel conductivities in CST after import. Numerical continuity at edges and seam boundaries depends on the specific tessellation and solid union settings, which should be retained alongside the geometry manifest for audit.

The packaged manifests record the generated meshes and geometry interpolation provenance. The process should be distinguished from direct import of a COMSOL field into a common coupled FEM system: COMSOL and CST remain separate numerical applications, with a mesh-based geometric transfer between them. This distinction matters because errors can enter in sampling, unit conversion, triangulation, material reassignment and coordinate-system orientation.

5.4 Unit Consistency and Model-Space Verification

The COMSOL displacement export states SI length units, whereas the STL generation and CST import use millimeter-scale geometry. Every coordinate transformation must therefore apply

$$\mathbf{r}'_{\mathrm{mm}} = 10^3\left(\mathbf{r}_{\mathrm{m}} + \mathbf{u}_{\mathrm{m}}\right). \tag{5.3}$$

A missed factor of 10^3 could make a microscopic deflection appear structural or could shrink the electromagnetic target below its intended size. The project file records an interpolated 35 m/s normal surface range of approximately -0.821 mm to $+0.0003$ mm, compatible in order of magnitude with the mesh-4 COMSOL peak displacement 0.833 mm. The agreement is an internal dimensional check, not proof of geometric interpolation convergence. Finite-thickness element orientation, component count, surface normals and conductivity attribution remain separate verification conditions.

5.5 Error Propagation and Model Hierarchy

A meaningful coupled uncertainty budget would consider $\Delta\mathbf{u}$ caused by FEM discretization, $\Delta\mathbf{r}_{\mathrm{IDW}}$ from interpolation, $\Delta\sigma_e$ from material uncertainty, $\Delta\boldsymbol{\theta}$ from mounting alignment, and ΔP_r from endpoint equipment uncertainty. As a local approximation,

$$\Delta P_r \approx \frac{\partial P_r}{\partial \mathbf{u}} \cdot \Delta\mathbf{u} + \frac{\partial P_r}{\partial \boldsymbol{\theta}} \cdot \Delta\boldsymbol{\theta} + \frac{\partial P_r}{\partial \sigma_e}\Delta\sigma_e + \cdots . \tag{5.4}$$

No probabilistic uncertainty distribution was calibrated for these variables. Consequently, the 75-case sweep explores chosen deterministic conditions rather than a statistically certified reliability envelope. This is especially important because the structural discretization error remains appreciable.

CHAPTER 6

Electromagnetic Simulation Campaign

6.1 Scattering Solver and Physical Representation

CST Studio Suite 2025 provides the high-frequency asymptotic SBR environment used for the electrically large conductor. The model involves finite-conductivity steel and zinc material layers, the central seam, and the displaced geometry constructed from the actual FEM field. SBR traces rays and estimates scattering according to high-frequency geometric and physical-optics principles. Its computational efficiency is useful for a $\sim 37\lambda$ aperture and a multi-case attitude sweep, but it is still an approximation; edge diffraction, detailed contact and possible local resonances are not independently confirmed by a separate full-wave integral-equation solution [21, 5, 2].

The archived master CST project is approximately 48 kB in packaged form, and its project status must not be conflated with the 75 full solver-run project directories excluded from the GitHub subset. The reproducibility evidence in the distributed package is strongest at the level of native far-field complex exports, accompanying angle case index and postprocessed RCS tables.

6.2 Incidence, Observation and Angular Grid

The incident wave uses a fixed reference geometry corresponding to the nominal 45° incidence/observation cut. The native CST field format contains frequency, incident azimuth/elevation, observation azimuth/elevation, and the real and imaginary components of E_ϕ and E_θ. For each wind case, the model evaluates two independent panel rotations on the set

$$\theta_x, \theta_y \in \{-0.2^\circ, -0.1^\circ, 0^\circ, +0.1^\circ, +0.2^\circ\}. \tag{6.1}$$

The full experimental design has $3 \times 5 \times 5 = 75$ unique wind/attitude configurations. Each is exported for TE and TM, yielding 150 angular-field records. These are *150 polarization-specific results*, not 150 statistically independent field measurements.

6.3 Extracting RCS from Native Complex Scattered Fields

The supplied parser selects the 5.5 GHz record at observation azimuth $\phi = 0^\circ$ and nominal receiver elevation $\theta = 45^\circ$, with a tolerance of 0.03° in elevation. Its four complex-field scalar values are used in

$$\widehat{\sigma}_b = 4\pi \left([\mathrm{Re}\, E_\phi]^2 + [\mathrm{Im}\, E_\phi]^2 + [\mathrm{Re}\, E_\theta]^2 + [\mathrm{Im}\, E_\theta]^2 \right). \tag{6.2}$$

Equation (6.2) incorporates the particular source-file convention of a unit-amplitude incident field and the solver's normalized scattered far field. The required distance normalization and field quantity definition must be preserved if raw records are regenerated under different export settings. The parser records both the fixed-receiver RCS and the maximum value over the exported angular cut. These are different observables; the fixed receiver direction is used for the link budget.

6.4 Actual Angular Response

Figure 6.1 plots the 35 m/s, zero-tilt native TE/TM complex-field cuts converted to dBsm using the documented parser expression. The lobe is localized around the desired observation region rather than being an omnidirectional response. The panel's large electrical aperture accounts for both appreciable specular scattering and sensitivity to small pointing errors.

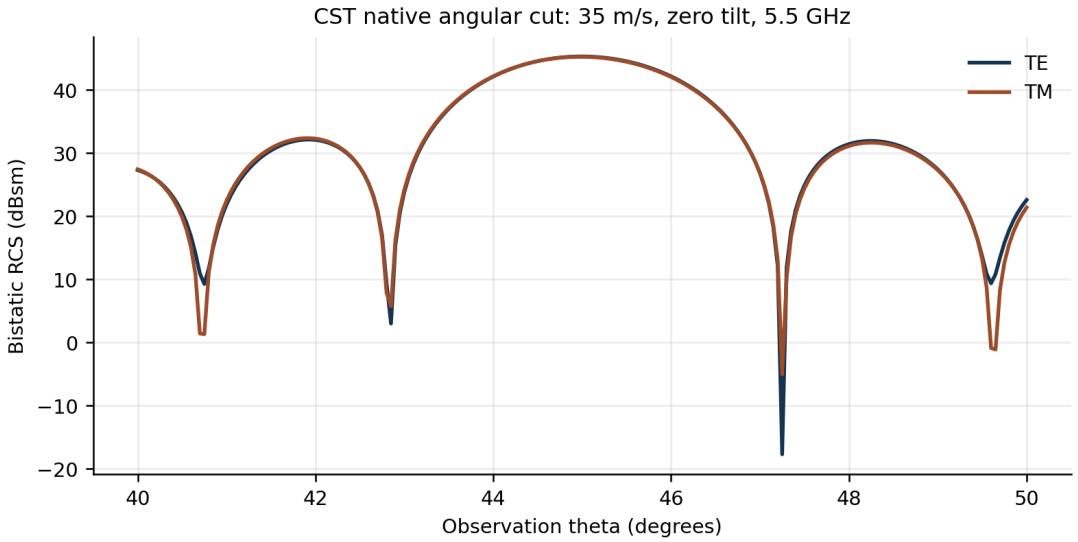


Figure 6.1: Bistatic RCS cut computed directly from original CST complex TE/TM field exports at 5.5 GHz, 35 m/s mesh-4 geometry and nominal alignment. It is a plot of solver data, not a conceptual curve.

6.5 Two-Axis Attitude Sensitivity

Figure 6.2 visualizes all 25 attitude samples for each polarization at 35 m/s. The tabulated worst cases do not coincide with perfect alignment. For TE, the least favorable point is $\theta_x = -0.2°, \theta_y = +0.2°$; for TM, it is $\theta_x = -0.2°, \theta_y = -0.2°$. This difference is consistent with polarization-dependent scattering and the chosen rotation conventions. A 5×5 discrete sweep cannot guarantee that the continuous worst angle is located exactly at a sampled grid point. An intermediate angle or installation tolerance outside the grid may yield a worse value.

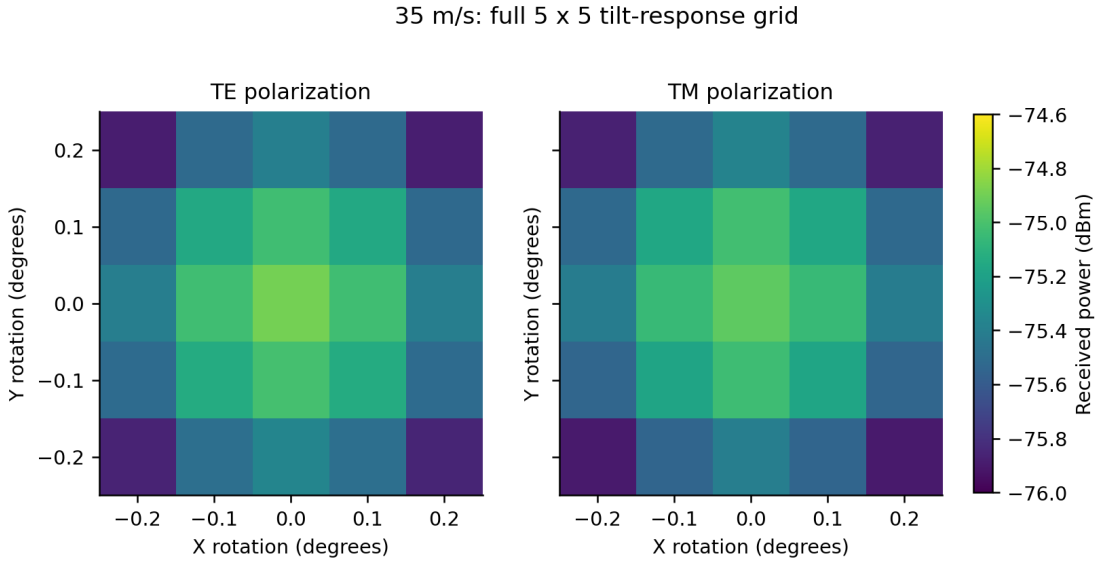


Figure 6.2: Directly tabulated CST/FEM 5×5 angular sensitivity for the 35 m/s condition; the plotted positions and received-power color values are from `LINK_CASES.csv`.

6.6 Polarization and Wind-Condition Results

Table 6.1 reproduces the six principal combinations from the archived results. For both polarizations the nominal received power remains close to −74.9 dBm; the worst sampled tilt makes the link about 1 dB weaker. The changes among wind cases are much smaller than the alignment penalty within this specific mesh-4 shape family. This comparison is conditional on the submitted structural fields and does not imply that stronger wind is generally negligible under a converged, time-varying deformation model.

Table 6.1: Native-CST-Derived Link Results for Each Wind/Polarization Group

Wind	Pol.	Cases	Aligned (dBm)	Worst (dBm)	Worst SNR (dB)
15	TE	25	-74.8904	-75.8834	18.1063
15	TM	25	-74.9059	-75.8975	18.0922
25	TE	25	-74.8924	-75.8839	18.1058
25	TM	25	-74.9055	-75.8983	18.0914
35	TE	25	-74.8946	-75.8858	18.1039
35	TM	25	-74.9470	-75.9011	18.0886

6.7 Data Integrity and Independent Numerical Checks

The final curated GitHub package retains 150 native TE/TM angular-field files of approximately 36 kB each. Their content supports recomputation of case-by-case RCS values rather than relying solely on summary tables. The CSV index identifies 75 solved geometrical cases. The original generated result package also supplies `RF_VALIDATION.json` with 150 expected and 150 successful polarization records. This is a provenance and consistency check; it is not experimental metrological traceability or independent verification by an entirely separate electromagnetic method.

CHAPTER 7

Bistatic Link Budget and Performance Screening

7.1 Nominal Link Assumptions

The incident path length and receiver path length are each 500 m. The transmitter-side input is specified as EIRP, so a separate transmit-antenna gain must not be added again. The receiving antenna is assigned 23 dBi gain in the desired scattered-wave direction. A further 6 dB loss is deducted collectively to account for unspecified link penalties. The channel is evaluated at a single 5.5 GHz frequency and 20 MHz receiver bandwidth. No actual terminal radiation pattern, feeder measurement or sensitivity curve is archived.

Table 7.1: Declared Link-Budget Parameters

Input	Assigned value
Center frequency	5.5 GHz
R_1, R_2	500 m each
EIRP	29 dBm
Receiving gain	23 dBi
Additional RF loss	6 dB
Noise bandwidth	20 MHz
Receiver noise figure	7 dB
Screening threshold	−77 dBm
Reference in archived worst-case CSV	−80 dBm

7.2 Step-by-Step Calculation for a Recorded Case

Take the 35 m/s aligned-TE result, with $\sigma_b = 45.315\,52$ dBsm. Equation (2.13) gives

$$\begin{aligned} P_r &= 29 + 23 + 45.31552 + 20\log_{10}(0.0545077) \\ &\quad - 30\log_{10}(4\pi) - 40\log_{10}(500) - 6 \\ &\approx -74.895\ \text{dBm}. \end{aligned} \tag{7.1}$$

The value agrees with the archived aligned result of −74.894 65 dBm to the precision of rounded inputs. Receiver noise from (2.14) is −93.9897 dBm, so SNR ≈ 19.095 dB. This numerical reconstruction makes the reported values inspectable and explains precisely why changing EIRP or receiver gain shifts the simulated received power one-for-one in dB.

7.3 Worst-Sampled-Link Result and Thresholds

The least favorable of the 150 recorded polarization-specific entries is −75.901 110 3 dBm, occurring for TM at 35 m/s and the angular pair $(-0.2°, -0.2°)$. Its thermal-noise-referenced SNR is 18.0886 dB. The margin above the −77 dBm project screening target is

$$M_{-77} = -75.9011103 - (-77) = 1.098\,889\,7\ \text{dB}. \tag{7.2}$$

A separate archived `WORST_CASES.csv` column measures margins relative to −80 dBm (about 4.099 dB in the worst case). These are two *different reference levels*, not contradictory solver outputs. This dissertation treats −77 dBm as the design-screening threshold and −80 dBm only as a secondary archival comparison; neither level has been verified as a measured receiver demodulation sensitivity.

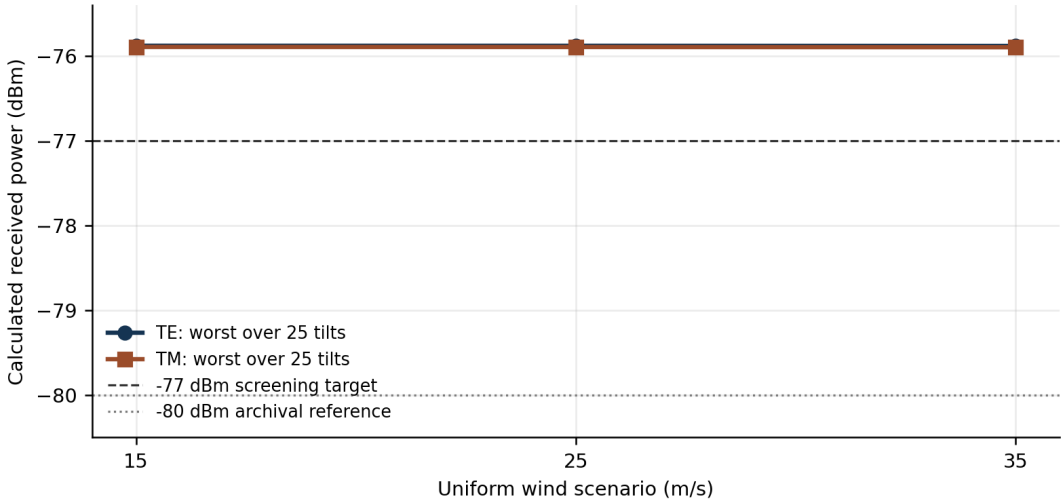


Figure 7.1: Worst calculated receiving power for TE and TM across each wind case, with both the project screening level (-77 dBm) and the separate archival -80 dBm reference made explicit. Source: genuine CST-derived link table.

7.4 Parametric Sensitivities

Because (2.12) contains $R_1^{-2}R_2^{-2}$, a range increase in either leg changes received power by $-20\log_{10}(R'/R)$ dB when all other quantities remain constant. Doubling *both* ranges therefore incurs 12.04 dB additional propagation loss, a much larger decrement than the present 1.10 dB screening margin. A 3 dB reduction in actual receiver gain, a 3 dB increase in unmodeled loss, or lower EIRP would each exceed this margin. More generally,

$$\Delta P_r = \Delta P_{\mathrm{EIRP}} + \Delta G_r + \Delta\sigma_{\mathrm{dBsm}} - \Delta L_{\mathrm{ext}} - 20\log_{10}\frac{R_1'}{R_1} - 20\log_{10}\frac{R_2'}{R_2}. \tag{7.3}$$

The result establishes that PASSBRIDGE's nominal gain is plausible within the assumed budget but does not have enough demonstrated reserve to absorb arbitrary real-world uncertainties.

7.5 Blocked Direct-Path Baseline

The no-reflector baseline is implemented as an explicitly blocked direct path with no alternative propagation contribution. In that mathematical topology the receiver receives no coherent desired signal, so the simulated no-link packet-error rate is one and useful throughput is zero. This is an *assumption-defined counterfactual*, not a field measurement proving that every geographical location has zero reflected, diffracted or scattered alternatives. Actual radio links may receive diffraction, vegetation scatter, building reflections or other multipath. Models based on ITU-R P.525 and P.530 require a properly specified path geometry and additional environmental inputs before availability can be inferred [7, 8].

CHAPTER 8

Digital Communication Model and MATLAB Results

8.1 Signal-Processing Objective

The link-budget result is a received-power estimate at a designated observation direction. To connect that estimate to a baseband information-transfer demonstration, the archived MATLAB simulation constructs a simple OFDM physical-layer model with uncoded QPSK and an additive Gaussian channel. Its role is limited but useful: it shows that, when the calculated SNR is supplied to this idealized receiver model, the transmitted bits can be demodulated with the finite-sample outcomes reported. It is not a full IEEE 802.11 modem or a statistical test of real-world channel availability [22, 23, 26].

8.2 OFDM Parameters and Modulation

Each symbol uses $N = 64$ IFFT points, with 52 occupied frequency bins $k \in \{-26, \ldots, -1, 1, \ldots, 26\}$, QPSK (2 bits per occupied carrier), and a cyclic prefix of $N_{\mathrm{CP}} = 16$ time samples. The 14 OFDM symbols per simulation frame carry

$$N_{\mathrm{bits/frame}} = 52 \times 14 \times 2 = 1456 \text{ bits}. \tag{8.1}$$

QPSK maps independent pairs of bits (b_I, b_Q) to the unit-energy constellation

$$s = \frac{(1 - 2b_I) + j(1 - 2b_Q)}{\sqrt{2}}, \qquad b_I, b_Q \in \{0, 1\}. \tag{8.2}$$

The OFDM time-domain samples are formed with a normalized inverse discrete Fourier transform:

$$x[n] = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}, \qquad 0 \le n < N, \tag{8.3}$$

with unoccupied tones set to zero. A 16-sample prefix copies the last 16 time-domain samples to the start of each symbol. In a frequency-flat AWGN channel no delayed multipath component is present; the CP is retained as a consistent waveform overhead assumption rather than as a measured mitigation of delay spread.

8.3 AWGN and Demodulation

The discrete complex baseband channel is

$$y[n] = x_{\mathrm{CP}}[n] + z[n], \qquad z[n] \sim \mathcal{CN}(0, \sigma_z^2), \tag{8.4}$$

with noise variance selected from the case-specific SNR and the signal power estimated on the transmitted CP-extended waveform. The receiver removes the prefix, computes a 64-point FFT, reads the 52 occupied bins and performs hard-sign decisions on the real and imaginary components. No fading-channel estimation, equalizer, time/frequency synchronization impairment, coding gain, interference, Doppler, retransmission, aggregation or medium access contention is implemented.

For coherent Gray-mapped QPSK over a conventional AWGN channel, an ideal theoretical baseline is

$$P_b = Q(\sqrt{2E_b/N_0}) = \tfrac{1}{2}\,\mathrm{erfc}(\sqrt{E_b/N_0}), \tag{8.5}$$

where proper accounting for bits per symbol, OFDM occupancy, CP overhead and any estimator normalization is required before equating an input sample-SNR number with E_b/N_0 [23, 24]. A reported zero sample BER is consistent with a low error probability at the chosen SNR; it is not a mathematical statement that P_b is identically zero.

8.4 Uncoded Net PHY Data Rate

For FFT sampling rate $f_s = 20\,\text{MHz}$ and no channel coding, the symbol-payload bit rate follows directly from the waveform definition:

$$R_{\text{payload}} = \frac{N_{\text{data}} \log_2 M \, f_s}{N + N_{\text{CP}}} = \frac{52 \times 2 \times 20 \times 10^6}{64 + 16} = 26\,\text{Mbit/s}. \tag{8.6}$$

The code multiplies this quantity by $\max(0, 1 - \widehat{\text{PER}})$ to report a simple packet-success-weighted figure. This quantity excludes Ethernet/IP headers, MAC overhead and channel access, forward error correction, pilots, guard carriers, ACK frames, retries and scheduling. It therefore has the meaning of a simplified uncoded PHY payload rate, not end-user Wi-Fi throughput or data-service assurance.

8.5 Monte Carlo Trial Configuration

The source code fixes its random generator to a documented seed and generates 200 packets for each of 12 reflector-present combinations: three winds, two polarizations and two attitude conditions (aligned and worst sampled). It adds one stipulated blocked-direct-path/no-reflector reference condition, yielding 13 CSV rows. Each reflector-present trial contains $200 \times 1456 = 291{,}200$ attempted payload bits. The no-reflector baseline uses random decisions in the absence of a desired signal and is labeled separately.

The sample BER estimator is

$$\widehat{P}_b = \frac{N_{\text{erroneous bits}}}{N_{\text{evaluated bits}}}, \tag{8.7}$$

while packet error rate is

$$\widehat{P}_{\text{pkt}} = \frac{N_{\text{packets with at least one bit error}}}{N_{\text{transmitted packets}}}. \tag{8.8}$$

Both estimators are conditioned on the simplified assumed channel. They do not incorporate confidence guarantees merely because their observed value is zero.

8.6 Results Across 13 Conditions

The archive records nominal 26 Mbit/s success-weighted rate, zero observed BER and zero observed PER for the 12 reflector-present cases. Their SNR values range from approximately 18.09 dB at worst alignment to approximately 19.10 dB when aligned. The no-reflector counterfactual is assigned a missing useful signal, resulting in zero useful bit rate, unit packet-error rate and approximately 0.4986 random-decision BER. Figures 8.1 and 8.2 present print-formatted plots of the original MATLAB-generated CSV values; the original MATLAB PNGs are preserved unaltered in the source archive.

8.7 Statistical Reading of Zero Packet Errors

If independent packets with stationary error probability p produce zero errors over n trials, a simple one-sided 95% exact binomial upper bound satisfies

$$(1 - p_{95})^n = 0.05, \qquad p_{95} = 1 - 0.05^{1/n}. \tag{8.9}$$

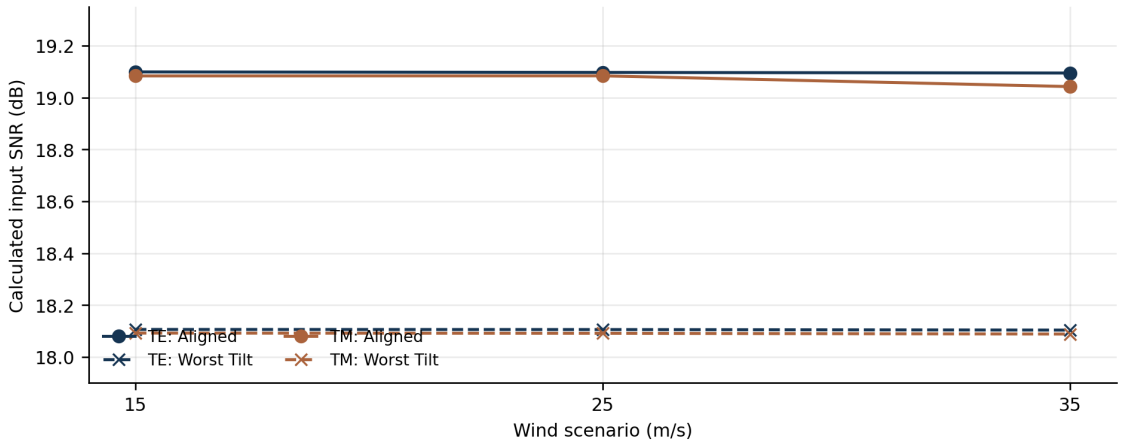


Figure 8.1: Print-formatted view of the archived MATLAB-derived SNR table: TE/TM aligned and worst-tilt inputs under three wind conditions. Numeric points reproduce project results; original MATLAB PNG is retained separately in the archive.

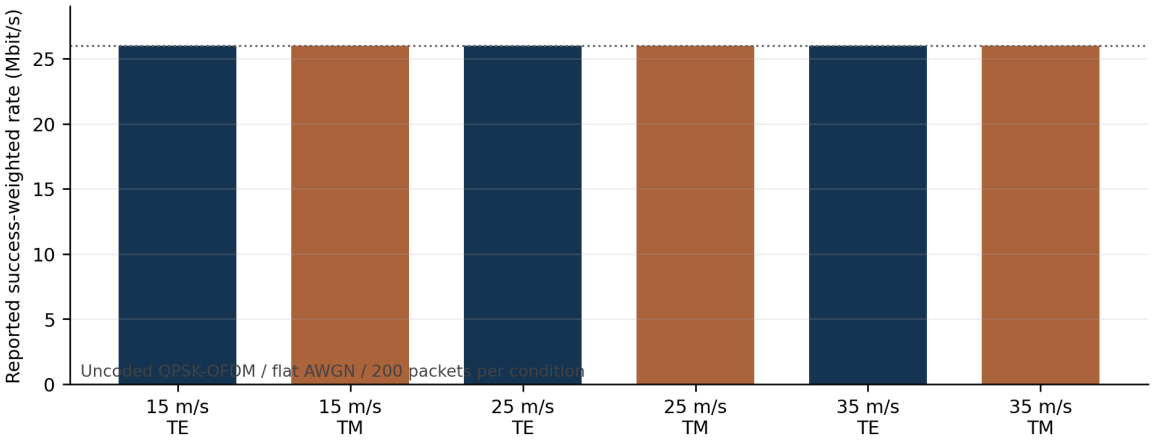


Figure 8.2: Print-formatted view of the original MATLAB result data: reported uncoded OFDM rates for the reflector-present conditions. Original MATLAB PNG is retained in the archive.

For $n = 200$, $p_{95} \approx 1.49\%$. Thus zero observed packet errors in each individual 200-packet condition does *not* demonstrate that its true packet-error probability is under 1% with 95% confidence. Combining separate conditions into a single sample pool would require a justified common error-probability model, which this experiment does not establish [25]. Furthermore, the idealized channel is precisely specified; a measured radio environment would introduce additional randomness and impairments.

8.8 Interpretation Relative to IEEE 802.11

IEEE Std 802.11-2024 defines multiple interoperable WLAN PHY and MAC specifications; the PASSBRIDGE QPSK-OFDM exercise only resembles the general multicarrier architecture and uses its own simplified occupancy, packet format and receiver. Neither protocol conformance nor a deployable 26 Mbit/s application-layer service follows from the MATLAB result [26]. The appropriate scientifically defensible description is *baseband feasibility under a fixed RCS-derived SNR and frequency-flat AWGN assumptions*.

CHAPTER 9

Integrated Results and Evidence Assessment

9.1 Performance Across the Modeling Chain

The completed simulation campaign is best read as a layered set of evidence. COMSOL provides quasi-static structural displacement samples for an idealized bonded frame. The displacement samples are mapped into a finite-thickness conductor description. CST generates polarization-resolved complex angular fields for three winds and 25 attitude samples. The bistatic link formula translates a fixed observation-angle RCS to terminal input power with independent equipment assumptions. MATLAB then tests a simplified waveform against a Gaussian noise floor derived from the power budget. This dependency graph is essential: downstream statistical metrics inherit modeling choices and uncertainties from upstream layers.

Table 9.1: What Each Evidence Layer Establishes

Stage	Computed evidence	Appropriate interpretation
Mechanical FEA	12 peak-value cases; three mesh-4 displacement fields	Conditional static responses; nonconvergent peaks
FEM-to-RF transfer	3 spatial deformation manifests and finite-thickness surface meshes	Traceable geometric projection of archived fields
CST SBR	75 solved attitudes; 150 native TE/TM complex cuts	High-frequency model scattering under specified illumination
Link budget	150 case-by-case power/SNR estimates	Deterministic receive-power projection for assumed radios
MATLAB	12 reflector cases plus one assumed null path	Idealized AWGN uncoded QPSK-OFDM finite-trial performance

9.2 Consistency with Aperture Theory

The principal electromagnetic sanity check is the numerical agreement between the gross physical-optics scaling law and the recorded nominal RCS. For the geometric aperture and 45° reference, (2.9) predicts approximately 45.3 dBsm. The CST nominal coupled result is about 45.30 dBsm, with modest TE/TM differences. This does not independently prove the boundary-value solution. It does, however, argue against errors of many orders of magnitude in area, wavelength or the RCS definition. A separate independent solver and calibrated field measurement would be stronger forms of validation [5, 19].

9.3 Power Threshold Versus Apparent Communications Success

The weakest link-model output, −75.901 dBm, exceeds the project's −77 dBm screen by only approximately 1.10 dB. This is a narrow deterministic margin. It can be compared with uncertainties not independently measured in the case study: effective endpoint EIRP, polarization mismatch, true incidence beam footprint, installation angle, wet coating and surface condition, multipath interference, receiver implementation losses, unmodeled atmospheric attenuation and wind-driven motion. A 1 dB alteration in any one independent dB-domain factor can absorb almost the entire available screening reserve. The computed passing result must therefore be quoted alongside the declared assumptions rather than independently.

9.4 Alignment Error Dominates Reported Wind Trend in This Dataset

The wind-conditioned aligned levels vary by only hundredths of a decibel across the three cases, whereas sampled angular misalignment typically changes the received power by nearly one decibel. In this particular experiment, the angular error is the dominant observed perturbation relative to the aligned reference. Because the mechanical model is not mesh converged and the angular grid is finite, this is a descriptive ranking of the *archived sampled model*, not a guarantee that real wind deflection is unimportant. A field implementation would prioritize accurate panel attitude, repeatable adjustment and stable anchorage; structural stiffness is still a prerequisite for preserving that attitude.

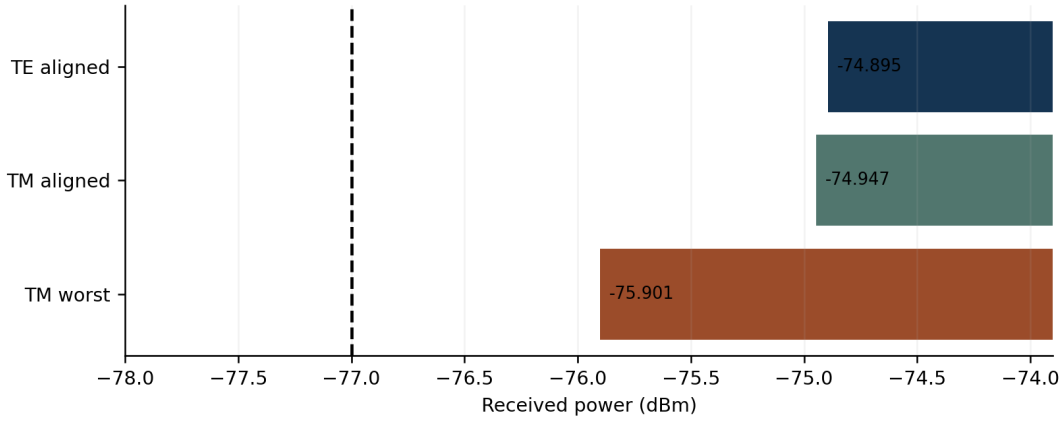


Figure 9.1: Three illustrative received-power points from the 35 m/s CST-derived results (aligned TE, aligned TM and worst sampled TM), plotted with the -77 dBm screening threshold.

9.5 Conductor Selection: Electromagnetic Adequacy Versus Mechanical Traceability

The zinc coating thickness estimate and microwave skin-depth calculation support the view that an intact Z275-coated surface can behave as an effective conductor at the chosen frequency. This is physically plausible and helps explain why an inexpensive steel sheet can serve as a passive scattering aperture. However, zinc thickness and steel conductivity are nominal, and DX51D's forming grade is not a complete structural specification. Consequently, electromagnetic viability of a coated sheet does not establish the allowable stress, corrosion life or certified wind resistance of a finished outdoor frame. The separate roles of electromagnetic surface properties, structural steel grades and joining methods must remain visible [9, 10, 12].

9.6 Key Numbers That Can Be Independently Recomputed

A technically competent reader can reconstruct the core narrative with relatively few primary quantities. The surface area is 3.996 m^2; the free-space wavelength is 54.508 mm; the Fresnel-to-far-field scale for a 2 m aperture is about 147 m; the Z275 nominal per-side coating is about 19.26 µm; and the simplified wind loads are 248.06, 689.06 and 1350.56 Pa. The archived worst-scattered-link power follows directly from (2.13) after insertion of the native RCS. The idealized payload rate follows from (8.6) with no fitted parameters. The numerical inconsistency is equally reconstructible: mesh-4 versus mesh-5 displacement differs by 64.76% when normalized to mesh 4. These checks form a coherent and auditable engineering result set.

CHAPTER 10

Uncertainty, Reproducibility and Research Validity

10.1 Numerical Verification Versus Physical Validation

Verification asks whether equations are solved and data are transferred correctly within the chosen numerical model. *Validation* asks whether that model adequately represents the intended physical system for its application. PASSBRIDGE provides evidence for solver execution and several internal consistency checks, including archived native field output, geometry manifests, linearly scaled wind responses and an algebraically reproducible link budget. It does not provide independent field measurements or manufacturing tolerances, so physical validation remains outside the data. This distinction is standard in computational science and engineering [19, 18].

10.2 Structural Discretization Error

The four available automatic COMSOL mesh levels do not demonstrate asymptotic convergence for the monitored peaks. The substantial fine-level increase may reflect treatment of thin geometry, local constraints, mesh-generated topology differences, stress singularities or other causes not isolated in the record. Regardless of mechanism, the available data cannot support a certified maximum displacement or design safety claim. Where exact spatial surface figure is a critical RF input, this uncertainty propagates directly to electromagnetic sensitivity.

10.3 High-Frequency Solver Approximation

The electrical size of the reflector supports SBR as a rational efficient first-order numerical choice. Nevertheless, a high-frequency approximation may omit some near-edge and junction physics and cannot substitute for a full-wave benchmark in every scenario. The available 150 exports demonstrate consistent behavior of the selected SBR model but are not 150 independent-method validations. Edge roughness, coating damage, interplate electrical discontinuity, contact resistance and detailed mounting geometry remain potential sources of discrepancy.

10.4 Single-Frequency and Incomplete Antenna Modeling

All native fields in the shared campaign are for 5.5 GHz, even though the assumed channel has nominal 20 MHz bandwidth. Wideband group delay, frequency-dependent RCS, polarization dispersion, frequency-selective fading and antenna pattern changes across an operating band were not supplied. The endpoint 29 dBm EIRP and 23 dBi receiver gain are fixed budget inputs, not measured or manufacturer-calibrated antenna patterns. The fixed-azimuth, fixed-elevation extraction represents one desired receive direction; it does not integrate actual antenna beam patterns over geometry.

10.5 Air-Interface Abstraction

The MATLAB analysis uses uncoded QPSK and ideal timing, a frequency-flat AWGN channel, no pilots or coding, and no protocol-layer packet exchange. It is particularly instructive as a *relative link-screening tool*: stronger or weaker simulated RCS changes calculated SNR, which influences the selected idealized baseband experiment. It cannot establish spectral regulatory compliance, cochannel-interference immunity, synchronization robustness, operational fade margin or 802.11 MAC throughput. All such claims would require a specified and verified radio implementation [26, 23].

10.6 Reproducibility Package and Provenance

The curated project archive distributes the principal structural model, the master CST model, model-generation and processing sources, native TE/TM far-field cuts, FEM displacement field tables, CSV summaries and the MATLAB figures. The original three-dimensional full solver caches and all 75 bulky CST project copies are not necessary for examining the numerical results in this dissertation. Re-running every stage on a new machine may nevertheless require licensed COMSOL 6.4, CST Studio Suite 2025 and MATLAB R2025b installations, correct local paths, compatible solver APIs and valid access to model-specific materials and geometries. The scripts are part of the reproducibility record; they are not evidence of manual data acquisition or an independently validated turnkey installation.

The supplied SHA-256 manifest enables integrity comparisons on downloaded files. Such cryptographic integrity supports chain-of-custody auditing of bytes, not physical correctness of any model. For software and numerical methods, a full independent reproduction would also require recording operating system, software build, numerical tolerances, solver settings, triangulation details and complete dependency versions.

10.7 Scope of a Credible Research Claim

It is warranted to state that the modeled G1 galvanized surface has near-specular bistatic RCS of roughly 45 dBsm at 5.5 GHz, that the 75-angle/wind CST campaign yielded a worst screened power near −75.90 dBm under the declared link budget, and that the recorded idealized OFDM trials decoded without sampled errors. It is not warranted to state that the physical product has been proved at 1 km, that the structure is certified for 35 m/s local wind action, or that a real 802.11 system will sustain 26 Mbit/s. The numerical result is substantial precisely because its assumptions and limitations are explicit.

CHAPTER 11

Implementation Feasibility and Engineering Roadmap

11.1 System-Level Architecture

The nominal deployed architecture would contain a directional active transmitter, passive reflector assembly and directional active receiver. The panel is an entirely passive scattering component and therefore requires no local RF electronics, amplifier, oscillator or backhaul interface. It may still require nontrivial civil works: a suitable site, stable mounting, corrosion protection, environmental monitoring, maintenance access, and proper antenna and reflector alignment. Thus zero operating power at the panel does not imply zero installation cost or zero lifetime intervention.

11.2 Alignment and Mechanical Tolerances

The observed sensitivity to $\pm 0.2^\circ$ angular changes suggests that precise orientation should be treated as a design-control variable. At a nominal 500 m path length, an angular pointing perturbation of 0.2° corresponds to a transverse direction change of roughly $R\tan(0.2^\circ) \approx 1.75$ m at the distant end. This geometric displacement is an illustrative pointing scale, not the actual beam displacement after diffraction and scattering from the finite aperture. Mount adjustment resolution, locking stiffness, surveying tolerances and orientation under wind must be specified for a prototype.

11.3 Mechanical Certification Inputs

A deployable engineering package would require a certified sheet material and framing specification, fabrication drawings, verified welded/bolted joints, foundation and pole analysis, a site-specific wind design basis, fatigue assessment when applicable, corrosion category and coating details, and a converged structural numerical model or independent hand-calculation validation. The existing idealized four fixed pads are a useful research boundary condition but not a complete structural connection design. Structural safety certification should be assigned only after a governing design standard and qualified engineering review have been applied [13, 9].

11.4 Electromagnetic Prototype Test Protocol

A physically meaningful RF validation would first measure the two terminal antenna patterns, EIRP, receiver noise figure and cable losses. The panel's actual area, surface roughness, gap, conductor continuity and orientation would be inspected. A controlled reference link would then measure received signal level at the fixed terminal positions with the panel aligned, deliberately misaligned and physically removed. Polarization-specific, frequency-swept received power and temporal fading would be recorded, with direct-path leakage and alternative reflections characterized instead of assumed absent. The test must distinguish scattered received power from direct coupling, nearby buildings, terrain diffraction and instrument calibration drift.

11.5 Communication-Quality Experiment

A hardware stage would employ a specified radio protocol, fixed channel plan and calibrated RF equipment. Useful outputs would include received power, error-vector magnitude, packet error rate with confidence intervals, negotiated modulation/coding rate, application-layer throughput, latency, channel occupancy, interference and long-term availability. Repeating the trial over varied weather and alignment would reveal whether the small deterministic simulated power margin is operationally adequate.

Without these measurements the MATLAB number is properly a baseband calculation, not demonstrated throughput.

11.6 Cost and Sustainability Framework

The capital-cost question cannot be answered responsibly from the simulation alone. A practical comparison should include reflective steel and coating, frame, machining/welding, transport, mounting, structural/civil work, alignment labor, site access, equipment at the endpoints, maintenance and eventual replacement. The lifecycle cost of an active relay would include transceiver hardware, enclosure, power, energy supply or solar-battery system, permits and service. The passive reflector may be attractive where local power or powered infrastructure is difficult, but cost advantage is conditional and not quantified by the present dataset.

CHAPTER 12

Conclusions

PASSBRIDGE was developed as a physically grounded numerical investigation of a large, passive metallic aperture intended to redirect a non-line-of-sight microwave path at 5.5 GHz. The study establishes an explicit engineering baseline: a 3.996 m^2 reflecting surface divided into two steel sheets, zinc coating modeled with finite conductivity, a supported frame under three uniform static wind actions, FEM-based deformation transfer, a 75-configuration SBR sweep, and a consistent bistatic receiver budget coupled to a simplified QPSK-OFDM system.

The native complex electromagnetic records yield a near-specular nominal RCS close to the physically expected 45.3 dBsm aperture scale. With two 500 m segments, 29 dBm EIRP, 23 dBi receiving gain and 6 dB assigned extra loss, the lowest archived receiving power is approximately −75.901 dBm. This clears the selected −77 dBm numerical screening threshold by approximately 1.10 dB; it is not a wide environmental reserve. The associated 20 MHz, 7 dB noise-figure model produces a worst SNR near 18.09 dB. The MATLAB experiment, using uncoded QPSK, 64-point OFDM and 52 occupied bins, corresponds to an idealized 26 Mbit/s gross payload-symbol rate and reported zero errors in a finite set of AWGN packet trials.

The structural results also disclose an important numerical condition: peak deflection changes from 0.293 61 mm at mesh level 5 to 0.833 23 mm at the refined mesh level 4 under 35 m/s, accompanied by low element-quality indicators. Therefore the exact deformed geometry, structural stress extrema and any certified mechanical margin cannot be declared mesh-independent from the archive. The system-level finding is accordingly a *conditional numerical feasibility demonstration*, supported by traceable native field data and independently recomputable equations, rather than a field-proven radio link or an approved outdoor structure.

The engineering importance of the work lies in its complete, inspectable connection between conductor selection, aperture geometry, structural perturbation, polarization-resolved scattering, terminal receiver power and digital baseband modeling. Its natural next stage, should physical deployment be pursued, is a calibrated prototype and site-specific structural and propagation validation. Within its stated scope, the dissertation provides a coherent quantitative foundation for that decision without requiring further repetitive solver runs merely to restate previously archived results.

CHAPTER A

Complete Modeling Parameter Ledger

This appendix consolidates inputs so the theoretical derivations and the archived simulation can be reviewed without reverse-engineering values from prose. Unless otherwise stated, electrical parameters belong to the single 5.5 GHz test; mechanical parameters belong to the uniform static-load reference conditions.

Table A.1: Parameter Ledger for PASSBRIDGE G1

Parameter	Value or representation
Frequency	5.5 GHz
Wavelength in vacuum	0.0545077 m
Target deflection	90 degrees
Reference incidence	45 degrees from surface normal
First propagation leg	500 m
Second propagation leg	500 m
Front aperture width / height	2.000 m / 2.000 m
Actual metal reflecting area	3.996 m^2
Panels	2 of 0.999 m x 2.000 m
Nominal steel thickness	1.2 mm
Nominal central seam	2 mm
Zinc designation	DX51D+Z275
Nominal zinc coating both sides	275 g/m^2 combined
Derived coating per face	19.26 μm nominal
RF steel conductivity	6.0×10^6 S/m, assumed
RF zinc conductivity	1.69×10^7 S/m, assumed
Relative magnetic permeability	1, assumed
Mechanical Young's modulus	210 GPa, assumed
Poisson ratio	0.30, assumed
Steel density	7850 kg/m^3, assumed
External perimeter support	four 50 x 50 x 3 mm RHS
Internal stiffeners	six 40 x 20 x 2 mm RHS
Rear support pads	four 100 x 100 x 8 mm
Mechanical idealization	linear static bonded solids, fixed pads
Air density	1.225 kg/m^3
Wind coefficient	1.8, assumed
Wind velocities	15 / 25 / 35 m/s
Uniform equivalent pressures	248.0625 / 689.0625 / 1350.5625 Pa
COMSOL automatic mesh sizes	7, 6, 5 and 4 (archive)
FEM displacement export coordinates	x, y, z, u, v, w, solid.disp in SI

Table A.1 (continued)

Parameter	Value or representation
Tilt angles, either axis	-0.2, -0.1, 0, 0.1, 0.2 degrees
SBR configurations	3 winds x 5 x 5 tilt pairs = 75
Polarizations	TE and TM, 150 native exports
EIRP	29 dBm
Receiving gain	23 dBi
Collective external loss	6 dB
Bandwidth	20 MHz
Noise figure	7 dB
Main screening level	-77 dBm
Archival comparative reference	-80 dBm
OFDM transform size	64 samples
Occupied carriers	52 (26 negative and 26 positive)
OFDM cyclic prefix	16 samples
OFDM frames	14 symbols / packet
QPSK payload bits / packet	1456
Simulation trials / condition	200 packets
Reflector-present AWGN conditions	3 winds x 2 polarizations x 2 alignments = 12
Total reported communication conditions	13 (including no-link null path)

CHAPTER B

Detailed RF and OFDM Numerical Data

The following tables are transcribed from the received-power and communication CSV files. All values are simulation results under the declared reference conditions, not experimental data. The case-by-case 75-row RF ledger appears in Table F.1 after the 13-condition MATLAB summary.

Table B.1: Recorded MATLAB Communication Conditions

Wind	Pol.	Condition	SNR (dB)	BER	PER	Rate (Mbit/s)
15	TE	Aligned	19.0993	0	0	26
15	TE	Worst tilt	18.1063	0	0	26
15	TM	Aligned	19.0838	0	0	26
15	TM	Worst tilt	18.0922	0	0	26
25	TE	Aligned	19.0973	0	0	26
25	TE	Worst tilt	18.1058	0	0	26
25	TM	Aligned	19.0842	0	0	26
25	TM	Worst tilt	18.0914	0	0	26
35	TE	Aligned	19.0951	0	0	26
35	TE	Worst tilt	18.1039	0	0	26
35	TM	Aligned	19.0427	0	0	26
35	TM	Worst tilt	18.0886	0	0	26
–	–	No reflector*	–	0.4986	1	0

*No-reflector entry is an assumed blocked-direct-path null reference and contains no received-signal SNR; no terrain blockage was physically measured.

CHAPTER C

Traceability of Native Files and Figures

C.1 Original Project Records

The principal source archive for this dissertation is `PASSBRIDGE_GITHUB_READY_20261027_015804.zip`. It includes `models/cst/PASSBRIDGE_FINAL_MASTER.cst`, `models/comsol/G1_MESH4_W35.mph`, 150 native CST complex-field TXT files under `data/cst_native`, and three deformation tables under `data/fem_displacement`. The structural and RF summaries are located at `results/fem` and `results/rf`, and the MATLAB result table plus original plots are located at `results/communication`. The corresponding scripts, including the COMSOL Java geometry and the CST and MATLAB workflow sources, are retained under `src/pipeline`. An archive-level SHA-256 manifest and publication audit accompany the package.

C.2 Classification of Thesis Figures

The image captions distinguish three provenance classes. Figures 8.1 and 8.2 are publication-style redraws of the original MATLAB-exported data; the source package also retains the original MATLAB PNG files. Figures 4.1 to 4.3, 6.1, 6.2, 7.1 and 9.1 are plots calculated from *the actual submitted numeric solver exports and CSV tables*; their visual style was standardized for publication but the numeric values were not generated synthetically. Figures 1.1 and 3.1 are dimensional model-based engineering schematics, explicitly not software screenshots or physical-prototype photographs. No actual photograph of an assembled PASSBRIDGE reflector was included in the source archive.

C.3 Recomputing Critical Results

To verify the key RF chain, read each TE and TM text file, choose frequency 5.5 GHz and the specified receive-angle sample, compute RCS from complex field components using (6.2), and substitute that RCS into (2.13). Evaluate the noise floor with (2.14); then compare with the summary columns `RCS_dBsm`, `Pr_dBm`, `Noise_dBm` and `SNR_dB`. To check structural values, compare the `MESH_QA.csv` extrema with `FEM_RESULTS.csv` and `REFINED_FEM_RESULTS.csv`. For the field images, select front-face nodes in the COMSOL text export with $z \approx 0$, plot $1000w$ in millimeters against the actual XY nodal coordinates and compare qualitative spatial structure with Figure 4.3. To reproduce the MATLAB rate, evaluate (8.6) with the stated fixed parameters. The original run uses a fixed MATLAB random seed, but repeatability on another host can still depend on implementation and version details.

CHAPTER D

Illustrative Numerical Cross-Checks

D.1 Skin-Depth Order of Magnitude

With $\sigma_{e,\mathrm{Zn}} = 1.69 \times 10^7$ S/m, $f = 5.5 \times 10^9$ Hz and $\mu = 4\pi \times 10^{-7}$ H/m, the skin-depth estimate from (2.3) gives $\delta \approx 1.65\,\mu$m. The coating-to-skin-depth ratio is therefore $t_{\mathrm{Zn}}/\delta \approx 11.7$. A real coating may contain oxide and composition variations that require direct electromagnetic material characterization for higher confidence.

D.2 Physical-Optics Aperture Check

For $\lambda \approx 0.054\,507\,7$ m, $A = 3.996\,\mathrm{m}^2$, and $\theta_i = 45°$, (2.9) gives $\sigma_{\mathrm{PO}} \approx 3.38 \times 10^4\,\mathrm{m}^2$, or $10\log_{10}\sigma_{\mathrm{PO}} \approx 45.3$ dBsm. A scattering cross section larger than geometric area is physically permissible because RCS characterizes directional concentration of the scattered field rather than intercepted physical area.

D.3 Noise and Confidence Check

At 20 MHz, $10\log_{10}(20 \times 10^6) \approx 73.0103$ dB, so $N \approx -174 + 73.0103 + 7 = -93.9897$ dBm. The least favorable recorded received power of -75.9011 dBm therefore corresponds to 18.0886 dB SNR. For 200 packet trials and zero observed packet errors, the one-sided 95% upper limit on an independent fixed packet-error probability is approximately 1.49%. Neither derived check requires additional CST or COMSOL execution.

CHAPTER E

Source Code and Environment Notes

The GitHub source contains `PASSBRIDGE_G1_FEM.java` to define and evaluate the COMSOL solid-mechanics problem; `FEMDEN_STL.py` to produce closed conductor mesh geometry from exported FEM displacements; `CST_FEM_RF.ps1` to control the CST SBR case sweep; `DOGRULA_RCS.py` to parse complex native field outputs and compute bistatic link metrics; `MATLAB_OFDM.m` for uncoded OFDM simulation; and `RAPORLA.py` for source-based summary reporting. Both manual engineering inspection of solver settings and code-level reproducibility are compatible with a valid computational methodology. The record does not support the claim that every calculation was performed exclusively by manual GUI operations, and this dissertation therefore avoids asserting such a history.

The models were developed using COMSOL Multiphysics 6.4 (export header build 6.4.0.429), CST Studio Suite 2025 and MATLAB R2025b. Reproducing numerical outcomes also requires verifying the actual licensed solver modules, software installation settings, matching material assignments, existing script paths and finite-difference/interpolation conventions. The project archive's publication audit should be reviewed before redistribution of any file with external dependencies or unreviewed usage rights.

CHAPTER F

Complete 75-Condition RF Ledger

The table contains every wind/tilt configuration with separate TE and TM fixed-receiver RCS and received-power values. Numerical values are drawn from `LINK_CASES.csv`, itself derived from the native CST angular-field files. The complete high-precision CSV is retained in the project archive; compact values here are rounded for typography. No measurements were undertaken.

Table F.1: All 75 Coupled CST Conditions at 5.5 GHz

Wind (m/s)	θ_x (deg)	θ_y (deg)	TE RCS (dBsm)	TM RCS (dBsm)	TE P_r (dBm)	TM P_r (dBm)
15	-0.2	-0.2	44.351	44.308	-75.855	-75.897
15	-0.2	-0.1	44.699	44.672	-75.507	-75.533
15	-0.2	+0.0	44.806	44.797	-75.400	-75.409
15	-0.2	+0.1	44.685	44.687	-75.521	-75.518
15	-0.2	+0.2	44.323	44.339	-75.883	-75.867
15	-0.1	-0.2	44.720	44.666	-75.486	-75.540
15	-0.1	-0.1	45.070	45.030	-75.136	-75.176
15	-0.1	+0.0	45.182	45.155	-75.024	-75.051
15	-0.1	+0.1	45.056	45.045	-75.150	-75.161
15	-0.1	+0.2	44.692	44.696	-75.514	-75.510
15	+0.0	-0.2	44.849	44.809	-75.357	-75.396
15	+0.0	-0.1	45.199	45.174	-75.007	-75.032
15	+0.0	+0.0	45.316	45.300	-74.890	-74.906
15	+0.0	+0.1	45.182	45.188	-75.024	-75.018
15	+0.0	+0.2	44.816	44.836	-75.390	-75.369
15	+0.1	-0.2	44.720	44.666	-75.486	-75.540
15	+0.1	-0.1	45.070	45.030	-75.136	-75.176
15	+0.1	+0.0	45.182	45.155	-75.024	-75.051
15	+0.1	+0.1	45.056	45.045	-75.150	-75.161
15	+0.1	+0.2	44.692	44.696	-75.513	-75.510
15	+0.2	-0.2	44.351	44.309	-75.855	-75.897
15	+0.2	-0.1	44.699	44.673	-75.507	-75.533
15	+0.2	+0.0	44.806	44.798	-75.400	-75.408
15	+0.2	+0.1	44.685	44.688	-75.521	-75.518
15	+0.2	+0.2	44.323	44.339	-75.883	-75.867
25	-0.2	-0.2	44.350	44.308	-75.856	-75.898
25	-0.2	-0.1	44.698	44.672	-75.508	-75.534
25	-0.2	+0.0	44.805	44.797	-75.401	-75.409
25	-0.2	+0.1	44.684	44.687	-75.522	-75.519

Table F.1 (continued)

Wind	θ_x	θ_y	TE RCS	TM RCS	TE P_r	TM P_r
25	-0.2	+0.2	44.322	44.338	-75.884	-75.868
25	-0.1	-0.2	44.718	44.665	-75.488	-75.541
25	-0.1	-0.1	45.068	45.029	-75.138	-75.177
25	-0.1	+0.0	45.179	45.154	-75.027	-75.052
25	-0.1	+0.1	45.055	45.044	-75.151	-75.162
25	-0.1	+0.2	44.691	44.695	-75.515	-75.511
25	+0.0	-0.2	44.847	44.809	-75.359	-75.397
25	+0.0	-0.1	45.197	45.173	-75.009	-75.033
25	+0.0	+0.0	45.314	45.300	-74.892	-74.906
25	+0.0	+0.1	45.181	45.187	-75.025	-75.019
25	+0.0	+0.2	44.815	44.836	-75.391	-75.370
25	+0.1	-0.2	44.719	44.665	-75.487	-75.541
25	+0.1	-0.1	45.069	45.029	-75.137	-75.177
25	+0.1	+0.0	45.180	45.154	-75.026	-75.052
25	+0.1	+0.1	45.055	45.044	-75.151	-75.162
25	+0.1	+0.2	44.692	44.695	-75.514	-75.511
25	+0.2	-0.2	44.351	44.308	-75.855	-75.898
25	+0.2	-0.1	44.699	44.672	-75.507	-75.534
25	+0.2	+0.0	44.806	44.797	-75.400	-75.409
25	+0.2	+0.1	44.685	44.687	-75.521	-75.518
25	+0.2	+0.2	44.323	44.339	-75.883	-75.867
35	-0.2	-0.2	44.348	44.305	-75.858	-75.901
35	-0.2	-0.1	44.696	44.669	-75.510	-75.537
35	-0.2	+0.0	44.803	44.794	-75.403	-75.412
35	-0.2	+0.1	44.682	44.684	-75.524	-75.522
35	-0.2	+0.2	44.320	44.336	-75.886	-75.870
35	-0.1	-0.2	44.716	44.662	-75.490	-75.544
35	-0.1	-0.1	45.065	45.026	-75.141	-75.180
35	-0.1	+0.0	45.175	45.151	-75.031	-75.055
35	-0.1	+0.1	45.051	45.041	-75.155	-75.165
35	-0.1	+0.2	44.688	44.692	-75.518	-75.514
35	+0.0	-0.2	44.845	44.806	-75.361	-75.400
35	+0.0	-0.1	45.195	45.171	-75.011	-75.035
35	+0.0	+0.0	45.311	45.259	-74.895	-74.947
35	+0.0	+0.1	45.179	45.185	-75.027	-75.021
35	+0.0	+0.2	44.812	44.833	-75.393	-75.372
35	+0.1	-0.2	44.716	44.662	-75.490	-75.544
35	+0.1	-0.1	45.066	45.026	-75.140	-75.180
35	+0.1	+0.0	45.176	45.152	-75.030	-75.054

Table F.1 (continued)

Wind	θ_x	θ_y	TE RCS	TM RCS	TE P_r	TM P_r
35	+0.1	+0.1	45.052	45.041	-75.154	-75.164
35	+0.1	+0.2	44.688	44.693	-75.517	-75.513
35	+0.2	-0.2	44.349	44.306	-75.857	-75.900
35	+0.2	-0.1	44.697	44.670	-75.509	-75.536
35	+0.2	+0.0	44.804	44.795	-75.402	-75.411
35	+0.2	+0.1	44.684	44.686	-75.522	-75.520
35	+0.2	+0.2	44.321	44.337	-75.885	-75.869

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